

An educational activity on climate change for secondary students in Vanuatu: Does gender influence performance?

Charles Pierce¹

¹ Bishop Grosseteste University, Lincoln, United Kingdom

© The Author(s) 2025. (Copyright notice)

Author Correspondence:

Charles Pierce

17 Fynd Street,

Goode Beach

Western Australia 6330

Australia

Email: charliepierce19@gmail.com

ORCID: 0000-0001-5717-4704

Abstract

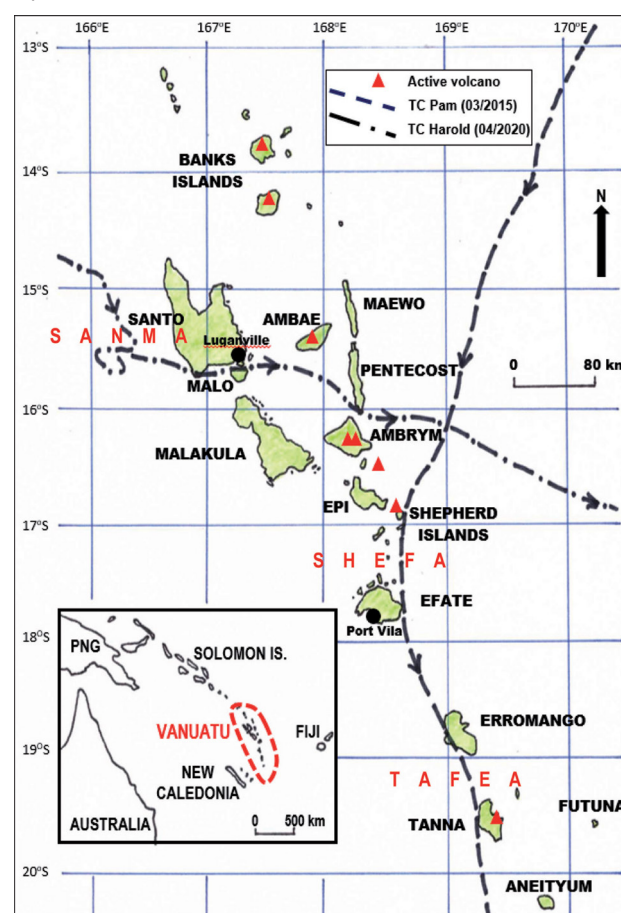
As part of wider research into the effectiveness of formal and traditional learning about climate and disaster resilience in the Pacific Island nation of Vanuatu, a mixed-method investigation was conducted during 2020 into the effectiveness of a pictorial resource depicting key elements of climate change education used in a discovery learning activity at junior secondary school level. Students completed a questionnaire before and after the intervention so that changes in their knowledge, attitudes, and behaviour could be measured. The quantitative data obtained was supplemented by qualitative information from teachers on the reasons for patterns observed. Results were compared across schools, age level, language of instruction, gender, and urban/rural location; this paper focuses on comparisons by gender obtained from a sample of 209 students, most of whom were in Years 9-10. While average overall scores for boys and girls, both before and after the intervention, were almost identical, girls performed better when their teacher was female and boys when their teacher was male. Also, girls had significantly higher scores than boys when the activity was conducted in urban schools, but not in rural schools. Reasons for this difference include a tendency for boys to be more easily distracted from their studies than girls in an urban environment.

Keywords: Vanuatu, climate change, gender, discovery learning, secondary education, junior secondary level, climate resilience, climate change adaptation, mixed methods

Vanuatu, an archipelago of over 80 volcanic islands in the south-west Pacific (Figure 1), is highly vulnerable to the impacts of climate change and disasters (Connell, 2013; Le Dé et al., 2018). Vanuatu's location in the tropical Pacific makes it susceptible to cyclones, floods, droughts, landslides, and sea-level rise, while its position on the "Ring of Fire" exposes it to earthquakes, volcanic eruptions and tsunamis (Pierce, 2019; Walshe & Nunn, 2012).

Figure 1

Map of Vanuatu



Note. Source: Author.

The 2021 World Risk Report, calculated for 181 countries on the basis of exposure and vulnerability (susceptibility, lack of coping capacities, and lack of adaptation capacities), identifies Vanuatu as the planet's most at-risk country to natural hazards, with an index of 47.73 on a range of 0-100 – considerably higher than that of the next two countries, the Solomon Islands (31.16) and Tonga (30.51; Bündnis Entwicklung Hilft, 2021).

In the 2020 national census, Vanuatu had a population of 300,019, of whom 66,753 (22%) were living in the two main urban areas of Port Vila and Luganville (Vanuatu National Statistics Office, 2021). In the same year, Vanuatu's Ministry of Education and Training (MOET) reported that total school enrolment was 96,829, of whom 23,943 were at secondary level (Years 7 to 14, with an official age range of 12 to 19; MOET, 2021).

The broader research project, conducted between October 2019 and November 2021, had the question "How effective is formal education on climate and disaster resilience in Vanuatu in terms of knowledge and skills gained, changes in attitude and behaviour and impacts on individuals and their communities?" (Pierce, 2023). This was applied to four academic levels – primary, junior secondary (Years 7 to 10), senior secondary (Years 11 to 13/14), and post-secondary. I investigated the impact of curricular materials and pedagogy on student learning about climate change and disasters, seeking to determine the level of resilience achieved. It must be stressed that this research in formal education was conducted in order to assess the impact of specific educational programmes or courses on the students involved, and not to assess the students themselves. The students served as agents for measuring the effectiveness of those courses or educational resources.

At junior secondary level, the existing Vanuatu curricula in Science and Social Science had minimal content on climate change or disaster risk reduction. New curricula were being developed but were not ready for use. The revision would mean that new topics related to climate change, disasters, and resilience would be inserted into syllabi for Social Science and Science, with the revisions linked to the United Nations Sustainable Development Goals 1-17, particularly Goal 13 on Climate Action (United Nations Department of Economic and Social Affairs, 2015).

There was, however, an educational resource for Pacific Island schools known as "Learning about Climate Change the Pacific Way" that had been developed between 2011 and 2013 in order to address the gap in learning materials

at junior secondary level on resilience to climate change and disasters in a Pacific Island context. It was one of the few resources available to schoolchildren and their teachers in Vanuatu in relation to resilience. It was produced under the auspices of the Secretariat of the Pacific Community (SPC) for use in five Pacific Island nations – Fiji, Kiribati, Samoa, Tonga, and Vanuatu – with funding provided by the SPC and Deutsche Gesellschaft für International Zusammenarbeit (GIZ) through their Coping with Climate Change in the Pacific Island Region Programme. Designed for Years 9 and 10, this "Climate Change Toolkit" comprises 16 large pictures for classroom display that deliver key messages relevant to climate change science, impacts in the Pacific, and options for mitigation and adaptation (SPC & GIZ, 2014). The images depict: the water cycle; climate globally and in the Pacific; El Niño and La Niña; causes of climate change; air-land-ocean interrelationships; the changing climate of Vanuatu; impacts of cyclones and droughts; mitigation and adaptation activities; adaptation strategies for cultivation, livestock, forestry, agro-forestry, and fishing; adaptation in urban areas; and an overview of all strategies on a Pacific island before and after adaptation. Figure 2 provides an example of one of these images.

Figure 2

Picture 11 of the Climate Change Toolkit and its Associated Questions



1. What actions can we take to reduce the levels of GHGs in the atmosphere? (Called "mitigation")
2. What actions can be taken to change our way of life so that it fits a climate that is warmer, with stronger cyclones and more droughts? (Called "adaptation")
3. What are the actions shown in the middle of the diagram?

Note. Source: SPC and GIZ (2014).

The Toolkit is accompanied by a comprehensive Teacher's Guide that provides background information and practical classroom activities applicable to both junior and senior levels of secondary education (SPC & GIZ, 2013). Its use was briefly critiqued by Walid (2017), but otherwise there have been no other attempts to evaluate its effectiveness. Walid's main criticisms were that:

Although the curriculum accounts for the power of tradition and uses pictures to communicate ideas directly, it omits traditional pedagogical methods of communication, such as story-telling. The project also does not cover all grades. Additionally, as this project relies heavily on top-down communication through government documents, it does not consider equally voices from the local communities.

(Walid, 2017, p. 277)

Sets of the Toolkit were distributed by Vanuatu's Curriculum Development Unit to junior secondary schools throughout the nation during 2013-2014, with short training courses for teachers on its deployment. By 2019, however, enquiries revealed that it had been largely forgotten. One reason given was the widespread destruction of learning materials as a result of Category 5 Tropical Cyclone Pam in March 2015; another was the frequent transfer of teachers between schools, with those who had received training moved into new posts and often asked to teach different subjects.

For my research into the effectiveness of learning about climate and disaster resilience at junior secondary level, I therefore set out to examine whether this resource does have intrinsic value in changing students' understanding of climate change and disasters, as well as attitudes and pro-environmental behaviours. At the same time, the aim was to increase awareness among new and existing teachers of the resource's existence, and how it might advance learning in the current absence of other materials on climate and disaster resilience for students at junior secondary level. I wanted to examine the impact of this educational resource on students in a variety of settings, as well as looking at variables that might influence its effectiveness: for example, language of instruction, urban or rural location, year level, and gender.

Literature Review

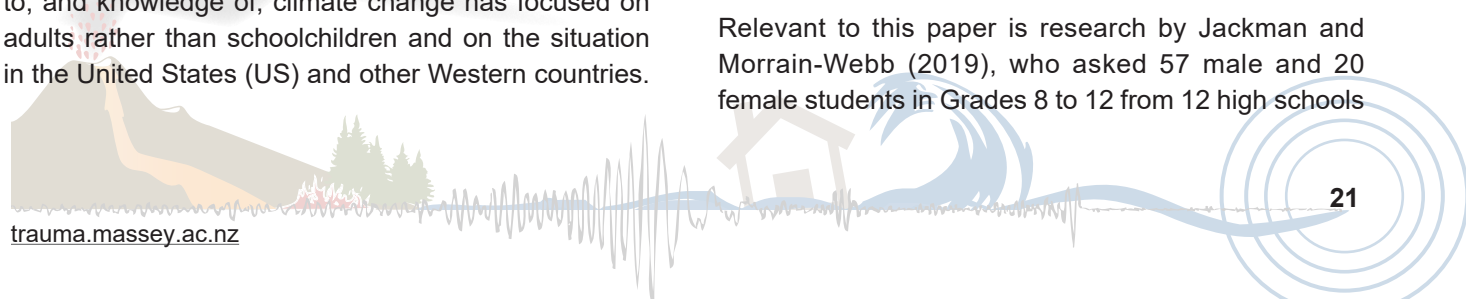
Gender differences among adults regarding climate change. Literature on gender differences in attitudes to, and knowledge of, climate change has focused on adults rather than schoolchildren and on the situation in the United States (US) and other Western countries.

For example, women are typically more likely than men to believe that climate change is happening (Hornsey et al., 2016; McCright et al., 2013), worry about its effects (McCright, 2010; McCright & Sundström, 2013), perceive more climate risks (Brody et al., 2008; Hamilton, 2011; van der Linden, 2015), express more knowledge about climate change (McCright, 2010), and perceive global warming as posing a threat within their lifetime (Hamilton, 2011). Women are also more likely than men in several economically wealthier countries (US, Canada, United Kingdom, Australia, and Italy) to endorse the view that people will need to make major changes in their lifestyles to reduce the effects of climate change. Women's greater environmental concerns are thought to result from their heightened risk perception (Bord & O'Connor, 1997; McCright, 2010; Xiao & McCright, 2012) and their greater propensity to show compassion and express an "ethic of care" (Zelezny et al., 2000, p. 445). Ballew et al. (2018) found that although similar proportions of men and women think that global warming is caused by humans, women more strongly believe that global warming will harm them personally and will harm people, plants, and animals as well as future generations. On the other hand, women were less likely than men to know the scientific facts about global warming and expressed more uncertainty about their knowledge.

Note that all the above studies relate to adult perceptions of climate change gained through public communications, as opposed to perceptions of school students gained in the classroom through climate change education.

Differences between boys and girls in academic performance at secondary school level. The influence of gender on the academic performance of school-age children is a subject of long-standing debate. Hyde (2004) proposed the "gender similarities hypothesis", arguing that boys and girls are more alike than they are different. On the other hand, research in both economically developed and developing countries indicates that since the end of the 20th century, girls are outperforming boys at secondary level (Hadjar et al., 2014; Jha & Kelleher, 2006; Nnamami & Oyibe, 2016; Ullah & Ullah, 2019; Van Houtte, 2004). In North Pentecost, Vanuatu, Mahuri (2019) found that certain socio-cultural practices jeopardize both participation (school attendance) and performance (academic achievement) of all schoolchildren, with boys impacted to a greater extent than girls.

Relevant to this paper is research by Jackman and Morrain-Webb (2019), who asked 57 male and 20 female students in Grades 8 to 12 from 12 high schools



in Trinidad and Tobago why they felt that girls were outperforming boys in high school. Among the reasons given, the most common was that females' greater success is due to the current societal thrust towards female empowerment and gender equity, which aids girls' academic achievement but can also immobilize boys. A second reason was the effects of negative influences in society that impact more on boys than on girls: gang violence, delinquency, criminal activity, illicit drug use, and abuse. A third reason was that male students focus on "extracurricular activity" (especially sport), rather than on schoolwork. A fourth reason related to a belief that girls are more goal-focused, emotionally mature, and have a better work ethic. Other factors mentioned were that the education system favoured females – linked to the "feminization of schooling" hypothesis (Figueroa, 2000; Joseph, 2016) – and that there is a lack of parental direction for boys.

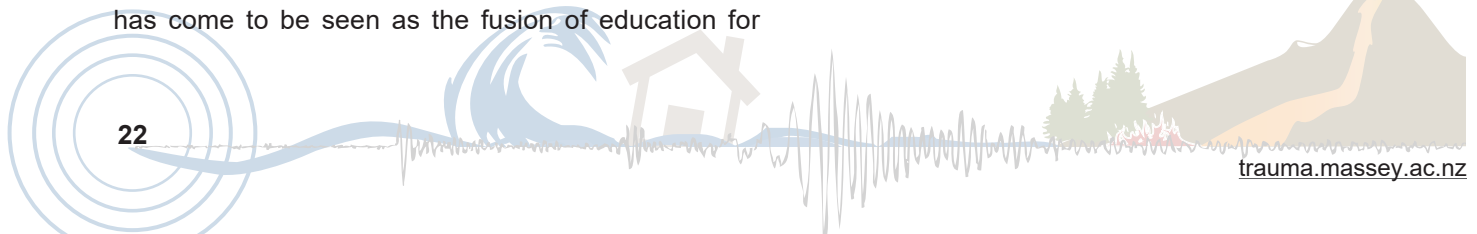
As we grow older, the reorganization of brain connections may explain why females generally mature faster in certain cognitive and emotional areas than males during childhood and adolescence (Lim et al., 2013). There appear to be general differences in brain development between male and female cohorts, and that modern pressures make it difficult for the young human brain to form optimal connections during adolescence, with boys being especially vulnerable (Bergland, 2013). These findings resonate with those of Brouwer et al. (2021), who investigated differences in brain age between males and females and found that although females and males show similar developmental ("aging") patterns, on average females pass through this development earlier, being ahead of males by at most a year. We may possibly infer that cognitive development, or the growth of a child's ability to think and reason (Stanford Medicine, 2024), occurs earlier in adolescent girls than it does for adolescent boys, and that this may impact on their faster academic progress in schools.

Differences between boys and girls regarding climate change education at secondary level. Most literature on climate change education appeared after the United Nations Educational Scientific and Cultural Organisation (UNESCO) launched the Decade of Education for Sustainable Development in 2010. Increasingly, greater focus has been placed on the role of adaptation in building resilience to impacts, with the aim of empowering learners to learn by doing and take action for sustainable development (Stevenson et al., 2017; UNESCO, 2015). Moreover, resilience education has come to be seen as the fusion of education for

climate change with education for disaster risk reduction (Kagawa & Selby, 2012), particularly for Pacific Islands, where climate change is viewed as a slow-acting disaster (Hemstock et al., 2017). It is within this context that the Toolkit is of relevance, since it has a particular emphasis on building adaptive capacity; according to the Teacher's Guide for this resource, the focus of learning is on "the effects of changes in air and sea-surface temperature, rainfall, sea-level rise and extreme weather events on environments, economies and people" and the need to "enhance individual and community skills to adapt to these changes" (SPC & GIZ, 2013, p.1).

Literature on the particular influence of gender on students' performance in climate change education at secondary school level is limited. Relevant to the present research is a study in Tanzania, where Kira and Komba (2015) collected information from 480 secondary school students in Morogoro region and found that boys were significantly more knowledgeable than girls in their understanding of climate change; they had greater awareness, for example, of the role played by carbon dioxide in global warming and of industries and automobiles as the most important contributors to greenhouse gas emissions.

This paper aims to address the gap in research on the role of gender in student performance in climate change education, particularly in terms of adaptation and resilience. We explore whether participation in the Toolkit activity by students in Vanuatu yields a differentiation in performance by gender. In this context, "performance" will be measured by scores achieved in a "before/after" diagnostic questionnaire for knowledge, attitudes, intended patterns of behaviour, and all three aspects together. This paper also seeks to determine whether teacher gender is a factor in educational success. For example, Muralidharan and Sheth (2016) found that male and female teachers of Grades 1-5 in Andhra Pradesh, India, were more effective in teaching students of their own gender, but that female teachers had greater effectiveness overall. However, Nnamani and Oyibe (2016), researching secondary students of Social Studies in Nigeria, discovered that male and female students taught by male teachers obtained higher mean scores than male and female students taught by female teachers.



Methods

Use of Quantitative Tools

This research used quantitative methods to determine the effectiveness of the Toolkit in changing students' knowledge, attitudes, and intended behaviour, and hence their level of resilience to climate change and disasters. A diagnostic questionnaire was designed for use before and after students undertook the Toolkit activity and this was to be completed by students at Year 9 and 10 level in a sample of classes selected from the 111 secondary schools in Vanuatu.

Selecting the Sample

This sample was not selected on a random basis. Instead, I relied on my long-standing contacts with teachers, especially those whom I had taught or trained over four decades of service in the country – teaching at secondary level, in tertiary level courses at USP's Emalus Campus in Port Vila, and at the Vanuatu Institute of Teacher Education (VITE). I identified possible English- and French-medium schools in urban and rural settings on several islands, then consulted with school principals and teachers to find classes whose teachers were willing to participate in the research. In this way, a sample of 19 classes was selected, with all students in those classes completing the same questionnaire both before and after undertaking the Toolkit activity. Although the focus was on Year 9-10 students, a few teachers felt that the activity would help students studying climate change in their Earth Science and Geography courses at senior secondary level, and asked to use it as a supplementary resource for those students. This was deemed acceptable as it could later enable a comparison of performance by year level. Questionnaires were completed by a total of 363 students from 19 classes at Year 9-12 level, of whom 296 (82%) were in Years 9 or 10.

Thus the sampling methods used for evaluating students' participation in resilience education through the Toolkit activity are complex. Schools and teachers were selected through non-probability sampling. The researcher used methods that fit the definitions of convenience and purposeful sampling (DeCarlo, 2018; Marshall, 1996), making deliberate choices to ensure participation from a range of schools in different settings. But once the schools and classes had been selected, then all students in each class were involved. All classes in Vanuatu schools are mixed, and at secondary level, females slightly outnumber males: in 2020, 12,287 females were enrolled in Years 7 to 14 as compared with 11,656 males (MOET, 2021). Those in Year 9 and above have already

passed a national examination at the end of Year 8, so that broadly speaking they are of similar academic ability, even though they vary in terms of family background, culture, and gender. In that sense they constitute a uniform population regardless of the school they attend, since in my experience, the Vanuatu school system is relatively equitable.

Investigating the Influence of Gender

The selection of schools and classes was made through purposeful sampling in order to capture a balance between urban and rural schools and between classes taught by female and male teachers, as well as similar overall numbers of boys and girls. Overall, data was disaggregated by gender for 112 girls and 97 boys from eight classes in five schools. The classes taught by female teachers had 74 female and 53 male students, while those taught by male teachers comprised 38 females and 44 males. Thus in terms of participation, classes with female teachers had higher proportions of females than males, while classes with male teachers had higher proportions of males than females. This may simply be co-incidence, or it may reflect my own observation from 35 years of teaching in Vanuatu that teenage boys at Years 9-10 level are more likely than girls to miss classes when their teacher is female. Note that 84% of Year 9 students are aged 14-17, and 85% of Year 10 students are aged 15-18 (MOET, 2022, calculated from Table 18).

Qualitative Tools

In order to explore potential reasons for differences in the performance of girls and boys, and between girls and boys in rural and urban settings, I conducted email interviews with seven of the teachers involved and three secondary school principals. In this way, a qualitative tool supplemented the quantitative data already obtained, triangulating information obtained from students with that gained from teachers.

Ensuring a Uniform Pedagogical Approach to the Toolkit Activity

In order to gain a realistic assessment of the Toolkit's effectiveness, I needed to ensure that the selected teachers would use the Toolkit activity in a uniform pedagogical manner. For this reason, I conducted a series of face-to-face training sessions in English and French between December 2019 and March 2020 for 17 teachers of Basic Science, Social Science, Geography, and Earth Science in each of six secondary schools, as well as for 13 teacher trainees graduating from the

VITE. I encouraged teachers to use the pictures in a discovery learning activity whereby the 16 images were pinned up in the classroom and students collaborated in pairs or small groups to visit each picture and answer its associated questions. This discovery learning activity might extend over 1 to 3 hour-long classroom sessions, depending on the students and the teacher, concluding with general whole-class discussion in which teacher and students would together explore and clarify key concepts. Figure 3 shows an example of one such teacher training session, while Figure 4 shows a group of students carrying out the activity.

Figure 3

Teachers of Year 10 Social Science at Nagavika College Participate in Training on the Climate Change Toolkit



Note. Here they are responding to questions on each picture as though they are students. Source: Author.

Figure 4

A Group of Students Discussing Picture 2



Note. Source: Social Science Teacher, Acacia Secondary School.

In order to assess changes in students' knowledge, attitudes, and behaviours as a result of this intervention, teachers would ask each individual student to complete the diagnostic questionnaire before undertaking the activity and then again at its conclusion (Figure 5), after the class had discussed the pictures with their teacher. Questions in the questionnaire were not the same as those accompanying each picture but attempted to draw out key aspects of learning generated by the Toolkit as a whole.

Figure 5

Individual Students Completing the "After" Questionnaire



Note. Source: Social Science Teacher, Pandanus Junior Secondary School

In all, 363 students from 19 classes in nine schools participated in the activity between March 2020 and March 2021. Implementation of the Toolkit activity was delayed by the declaration of a State of Emergency (SOE) in Vanuatu in April 2020 due to the impacts of Category 5 Cyclone Harold and the start of the COVID-19 pandemic. Restrictions on international travel meant that once I left Vanuatu in March 2020, I was unable to return until mid-2022. Progress was therefore monitored through virtual means. Completed "before" and "after" questionnaires were transmitted electronically, and interviews with teachers conducted through emails, texts, or phone.

Data Collection and Analysis

A mixed methods approach to the research was followed, with quantitative analysis of students' responses supported by qualitative data obtained from class teachers and school principals.

For the quantitative data, each class teacher asked his/her students to complete the diagnostic questionnaire before and after the intervention, then send all completed forms to the researcher. Questionnaires were marked according to the scheme shown in the Appendix, then the results for each class were tabulated. The diagnostic questionnaire comprises 20 agree/disagree (A/D) statements, each worth one point, and seven short-answer questions, each marked on a four-point scale according to levels of displayed competence. In marking the A/D statements, +1 was awarded for a correct answer, -1 for an incorrect answer and 0 for "Don't know", with this scale adopted in order to differentiate correct from incorrect responses. An exception was made for statement 18, where a respondent was not penalized for disagreeing to participate in a student strike for climate change because there are other ways in which concern can be expressed. For the longer answers required in questions 21 to 27, marking followed descriptors at four

levels (accurate, satisfactory, limited, and irrelevant/no answer).

For purposes of analysis, items 1-16 and 21-26 in the questionnaire were considered as indicators of knowledge, items 17-20 of attitudes/opinions, and item 27 of intended behaviour. Statement 19 was considered as an attitude as it reflects a student's level of confidence in his/her own ability. Scores can range from -1.0 to +1.0 for items 1 to 20, and from 0.0 to +5.0 for items 21 to 27. The higher the score, the higher the degree of resilience implied and so the better the performance.

For purposes of confidentiality, schools were given anonymous names that correspond to common tree species in Vanuatu.

Results

Comparison of Male and Female Scores: Averages for all Items, Knowledge, Attitudes, and Behaviour

Table 1 and Figure 6 show a comparison of average scores for females and males before and after the intervention. For all items, scores were almost identical: 0.49 for both genders before the activity and 0.81 for females and 0.80 for males afterwards.

For knowledge items, there was a similar pattern, with girls attaining average scores of 0.44 beforehand and 0.72 afterwards, with an increase of 0.28, while the comparable scores for boys were 0.45 and 0.71 with an increase of 0.27. For attitudes, females had a higher average score than males before the intervention (0.58 to 0.48), but a lower score afterwards (0.73 to 0.78), so that the improvement for females was half that for males (0.15 to 0.31). In terms of behaviour, males had a higher score than females beforehand (1.42 to 1.25), and a lower score afterwards (2.69 to 3.08), with a lower degree of improvement (1.27 to 1.83). Note that the scores for behaviour are based on one item only (Q27), which has a mark range of 0 to 5.

If, however, the average scores by gender are disaggregated into responses to the 20 A/D items and responses to the seven short-answer questions (Figure 7), a different pattern emerges for the knowledge items. Scores for the A/D items were slightly higher for boys than girls before the intervention (0.38 to 0.35) and afterwards (0.56 to 0.54), while it was the reverse for the short answer questions, with higher scores for girls beforehand (0.72 to 0.62) and again afterwards (1.16 to 1.10). The Toolkit seems to have improved girls' performance in the short answer questions for both

knowledge and behaviour to a greater extent than for boys, meaning that an understanding of climate change and its impacts and solutions had increased more for girls than it had for boys.

Figure 6

Comparison of Average Scores for Females and Males Before and After the Intervention

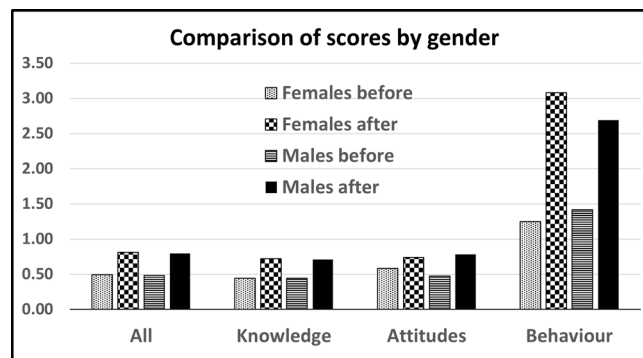
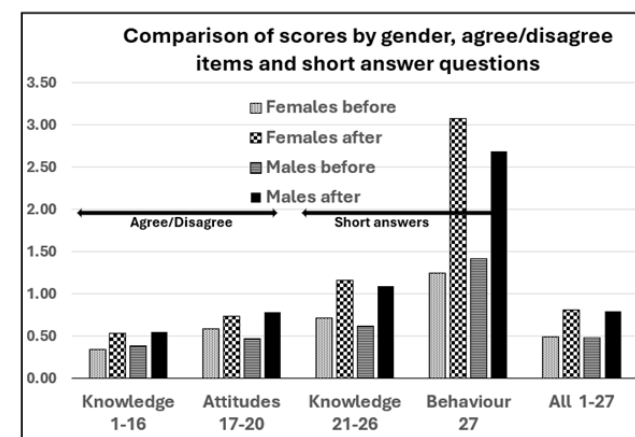


Figure 7

Comparison of Average Scores for Females and Males Before and After the Intervention, Differentiated According to Agree/Disagree Items and Short Answer Questions



Comparison of Male and Female Scores for Individual Items

Performance by gender can also be considered on the basis of individual items. Table 1 shows that for the A/D items, exposure to the Toolkit has resulted in boys outperforming girls in nine of the 16 items (Qs 2, 3, 4, 5, 9, 10, 12, 15, 16) that relate to climate science, fishing, and Vanuatu's vulnerability to disasters. Girls outperformed boys in Qs 1, 6, 7, 8, 11, 13, and 14, which refer to the effect of El Niño and definitions of weather, the greenhouse effect, adaptation, and sustainable agriculture. With the short answer questions, the intervention resulted in girls clearly outperforming boys in Qs 21, 26, and 27, which apply to the definition and

Table 1*Comparison of Average Scores for Females and Males Before and After the Intervention*

Statement / Question no.	Females (n = 112)			Males (n = 97)		
	Before Av. score	After Av. score	Change	Before Av. score	After Av. score	Change
1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.	0.94	0.92	-0.02	0.80	0.82	0.02
2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.	0.47	0.53	0.05	0.38	0.65	0.27
3. Our climate changes because of both natural and human factors	0.63	0.85	0.22	0.63	0.92	0.29
4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.	-0.30	-0.30	0.00	-0.04	-0.16	-0.12
5. Climate variability means that the climate of a place may change from year to year.	0.25	0.54	0.29	0.23	0.64	0.41
6. During an El Niño period, Vanuatu experiences droughts.	0.48	0.76	0.28	0.69	0.68	-0.01
7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.	0.30	0.76	0.46	0.33	0.64	0.31
8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.	0.27	0.78	0.51	0.25	0.48	0.24
9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.	0.16	0.43	0.27	0.18	0.46	0.29
10. Carbon dioxide is absorbed by forests and the oceans	0.29	0.45	0.16	0.38	0.77	0.39
11. Coral reefs are being damaged by warmer temperatures and ocean acidification.	0.39	0.68	0.29	0.46	0.66	0.20
12. Vanuatu's greatest climatic dangers are cyclones and droughts.	0.86	0.89	0.04	0.70	0.91	0.21
13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.	-0.27	-0.38	-0.12	0.00	-0.54	-0.54
14. Using compost is a sustainable form of gardening.	0.72	0.92	0.20	0.56	0.80	0.25
15. Catching fish with nets that have very small holes is a sustainable form of fishing.	-0.05	0.10	0.15	-0.01	0.39	0.40
16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.	0.41	0.71	0.30	0.55	0.76	0.22
17. I am worried that climate change will bring great dangers to the world in the future. *	0.63	0.88	0.24	0.42	0.81	0.39
18. I want to join a student strike to show my concern about climate change.*	0.69	0.70	0.01	0.71	0.81	0.10
19. I know a lot about the causes and impacts of climate change.*	0.20	0.44	0.24	0.11	0.61	0.49
20. I want to help my community prepare for climate change and disasters.*	0.82	0.93	0.11	0.66	0.90	0.24
21. What does "climate change" mean?	0.57	0.91	0.34	0.27	0.67	0.40
22. What is the enhanced greenhouse effect?	0.59	0.88	0.29	0.47	0.86	0.38
23. What are 3 human activities putting extra GHGs into the atmosphere?	1.47	2.08	0.61	1.16	2.08	0.92
24. What causes ocean acidification?	0.35	0.73	0.38	0.41	0.72	0.31
25. Why is forestry important for mitigating climate change?	0.66	1.15	0.49	0.72	1.21	0.48
26. What are some of the main impacts of climate change in Vanuatu?	0.58	1.46	0.88	0.67	1.23	0.56
27. What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters?	1.25	3.08	1.83	1.42	2.69	1.27
Knowledge	0.44	0.72	0.28	0.45	0.71	0.27
Attitudes	0.58	0.73	0.15	0.48	0.78	0.31
Behaviour	1.25	3.08	1.83	1.42	2.69	1.27
All	0.49	0.81	0.32	0.49	0.80	0.31

Note. * = Attitude questions; Item 27 = Behaviour question; all other items = Knowledge questions.

impacts of climate change, and behaviours to prepare for climate change and disasters. Girls slightly outperformed boys in Qs 22 and 24, concerning knowledge of the greenhouse effect and ocean acidification. Girls and boys achieved equal scores in Q 23, on anthropogenic causes of climate change, while boys outperformed girls in Q 25, relating to knowledge of how forests mitigate climate change.

Teacher Gender and Student Performance

Turning now to the question of whether the gender of the teacher affects student performance, Table 2 shows results for students in mixed classes taught by female teachers and those taught by male teachers. Average scores for all items before students participated in the Toolkit activity were significantly higher in classes taught by female teachers (0.54) than those taught by male teachers (0.41). However, participation in the activity enabled students taught by male teachers to attain almost the same score (0.79) as those taught by female teachers (0.81). The same pattern can be observed for performance in knowledge items, with classes taught by female teachers having initial scores of 0.50 as compared with 0.36 for classes with male teachers, and then the intervention bringing the scores for male-taught classes to almost the same level (0.71) as that for female-taught classes (0.72). For attitudes, classes taught by female teachers scored more highly before the start of the activity (0.58 to 0.46) but saw a smaller increase (0.73 to 0.80). In the one short answer question on behaviour, male-taught classes had a higher score at the start (1.38 to 1.30) but showed a smaller improvement (2.71 to 3.02), implying that in this respect, female teachers were more effective than males in helping their students become aware of a greater range of adaptation methods.

For individual items, participation in the intervention meant that classes taught by male teachers achieved much higher scores than those taught by females in knowledge items 1, 2, 6, 7, 11, 21, and 24, as well as in attitude items 17, 18, and 20. However, female-taught classes had significantly higher scores in attitude item 19 (confidence in knowledge of climate change), as well as in knowledge items 22, 23, 25, and 26 and in behaviour item 27.

A further question was investigated: whether the relationship between performance and the gender of the student depends on the gender of the teacher. Table 3 shows male and female students' performance in the Toolkit activity by gender of the teacher.

The most salient finding relates to students' average scores for all items before and after participating in the Toolkit activity. When a class had a female teacher, girls achieved significantly higher overall scores than boys, both before and after the activity: 0.58 and 0.86 for girls compared with 0.49 and 0.74 for boys. When the class had a male teacher, boys achieved significantly higher overall scores than girls, both before and after the activity: 0.72 and 0.86 for boys compared with 0.34 and 0.72 for girls. Thus, there is a certain symmetry about the results, with the pattern for male teachers being the mirror image of the pattern for female teachers; the "after" average score for female students taught by female teachers (0.86) was identical to the "after" average score for male students taught by male teachers.

Independent samples *t*-tests showed that when mixed classes taught by female teachers carried out the Toolkit activity, female students achieved significantly higher "after" scores than male students, $t(26) = 2.63, p < .05$. Similarly, when mixed classes taught by male teachers carried out the Toolkit activity, male students achieved significantly higher "after" scores than female students, $t(26) = 3.52, p < .01$.

We can infer that in the context of the climate change activity carried out in Vanuatu, a teacher's gender does have an influence on student achievement, with both male and female teachers being more effective at teaching students of their own gender.

Differences in Performance by Gender Between Students in Urban and Rural Schools

A further question on performance by gender is whether there was a difference between students in urban and rural schools. Of the five schools in the sample, two are located in the capital city of Port Vila, with a total of 75 females and 44 males. The remainder, located in rural areas, have a total of 37 females and 53 males. In all, the sample of 209 comprises 119 students in urban schools and 90 students in rural schools.

Table 4 shows before and after scores for all females and males in urban schools with before and after scores for all females and males in rural schools. There is a clear pattern in the overall performance; in urban schools, female students outperformed male students in average scores both before the intervention (0.54 compared to 0.47) and afterwards (0.84 to 0.76), while in rural schools, males outperformed females beforehand (0.50 to 0.40) and afterwards (0.83 to 0.76).

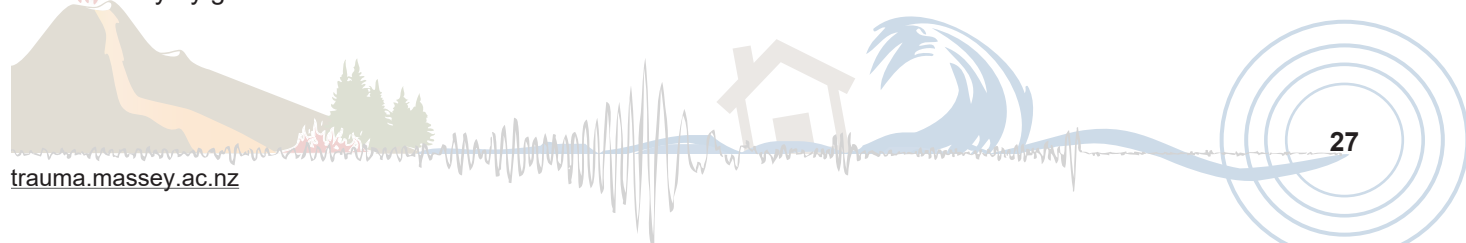


Table 2*Comparison of Average Scores for 209 Students Taught by Female and Male Teachers*

Statement / Question no.	Taught by female teachers (n = 127)			Taught by male teachers (n = 97)		
	Before Av. score	After Av. score	Change	Before Av. score	After Av. score	Change
1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.	0.91	0.80	-0.11	0.83	1.00	0.17
2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.	0.36	0.45	0.09	0.54	0.79	0.26
3. Our climate changes because of both natural and human factors	0.70	0.86	0.16	0.51	0.91	0.40
4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.	-0.17	-0.28	-0.10	-0.20	-0.18	0.01
5. Climate variability means that the climate of a place may change from year to year.	0.30	0.57	0.28	0.15	0.61	0.46
6. During an El Niño period, Vanuatu experiences droughts.	0.69	0.69	0.00	0.40	0.77	0.37
7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.	0.28	0.56	0.28	0.37	0.93	0.56
8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.	0.21	0.65	0.44	0.33	0.62	0.29
9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.	0.25	0.44	0.19	0.04	0.45	0.41
10. Carbon dioxide is absorbed by forests and the oceans	0.30	0.58	0.28	0.38	0.62	0.24
11. Coral reefs are being damaged by warmer temperatures and ocean acidification.	0.55	0.61	0.06	0.23	0.77	0.54
12. Vanuatu's greatest climatic dangers are cyclones and droughts.	0.89	0.89	0.00	0.62	0.91	0.29
13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.	-0.14	-0.34	-0.20	-0.15	-0.63	-0.49
14. Using compost is a sustainable form of gardening.	0.76	0.87	0.11	0.46	0.85	0.39
15. Catching fish with nets that have very small holes is a sustainable form of fishing.	0.06	0.26	0.20	-0.18	0.20	0.38
16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.	0.51	0.76	0.25	0.41	0.70	0.28
17. I am worried that climate change will bring great dangers to the world in the future. *	0.55	0.80	0.24	0.51	0.93	0.41
18. I want to join a student strike to show my concern about climate change.*	0.74	0.67	-0.07	0.63	0.88	0.24
19. I know a lot about the causes and impacts of climate change.*	0.21	0.60	0.39	0.07	0.39	0.32
20. I want to help my community prepare for climate change and disasters.*	0.82	0.86	0.04	0.63	1.00	0.37
21. What does "climate change" mean?	0.46	0.76	0.29	0.38	0.87	0.49
22. What is the enhanced greenhouse effect?	0.60	0.91	0.31	0.44	0.80	0.37
23. What are 3 human activities putting extra GHGs into the atmosphere?	1.71	2.39	0.69	0.74	1.60	0.85
24. What causes ocean acidification?	0.35	0.69	0.35	0.43	0.78	0.35
25. Why is forestry important for mitigating climate change?	0.74	1.25	0.51	0.61	1.06	0.45

	Before Av. score	After Av. score	Change	Before Av. score	After Av. score	Change
26. What are some of the main impacts of climate change in Vanuatu?	0.67	1.52	0.85	0.55	1.10	0.55
27. What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters?	1.30	3.02	1.72	1.38	2.71	1.33
Knowledge	0.50	0.72	0.22	0.36	0.71	0.35
Attitudes	0.58	0.73	0.15	0.46	0.80	0.34
Behaviour	1.30	3.02	1.72	1.38	2.71	1.33
All	0.54	0.81	0.27	0.41	0.79	0.38

Note. * = Attitude questions; Item 27 = Behaviour question; all other items = Knowledge

Table 3

Comparison of Average Scores for Female and Male Students Taught by Female Teachers with Those Taught by Male Teachers

	Female teachers				Male teachers			
	Female students (n = 74)		Male students (n = 53)		Female students (n = 38)		Male students (n = 44)	
	Before	After	Before	After	Before	After	Before	After
Knowledge (1-16)	0.4	0.55	0.42	0.49	0.25	0.53	0.34	0.63
Attitudes (17-20)	0.67	0.73	0.46	0.73	0.42	0.74	0.5	0.85
Knowledge (21-26)	0.86	1.35	0.61	1.12	0.4	0.92	0.63	1.14
Behaviour (27)	1.36	3.35	1.21	2.57	1.13	2.56	1.6	2.84
All	0.58	0.86	0.49	0.74	0.34	0.72	0.48	0.86

Table 4

Average Scores for Female and Male Students in Urban and Rural Schools Before and After Participating in the Toolkit Activity

	Urban schools				Rural schools			
	Female students (n = 75)		Male students (n = 44)		Female students (n = 37)		Male students (n = 53)	
	Before	After	Before	After	Before	After	Before	After
Knowledge (1-16)	0.37	0.53	0.39	0.49	0.29	0.55	0.37	0.61
Attitudes (17-20)	0.64	0.68	0.45	0.64	0.48	0.84	0.50	0.90
Knowledge (21-26)	0.78	1.32	0.57	1.19	0.54	0.97	0.66	1.07
Behaviour (27)	1.43	3.41	1.11	2.89	0.89	2.41	1.68	2.53
All	0.54	0.84	0.47	0.76	0.40	0.76	0.50	0.83

A similar pattern is observable in the results for knowledge (A/D items and short answer questions), attitudes, and behaviour. In all categories, and both before and after the intervention, girls outperformed boys in urban schools, but boys outperformed girls in rural schools.

Independent samples *t*-tests showed that there is a significant difference between average “after” scores of female and male students in urban schools, $t(26) = 2.23$, $p < .05$. However, the difference between average “after” scores of female and male students in rural schools was not statistically significant.

Seeking Reasons for the Difference in Performance by Gender Between Urban and Rural Schools

To explain why females scored significantly higher scores than males when the CC Toolkit activity was conducted in urban schools, but not in rural schools, two reasons can be suggested. Firstly, on purely statistical grounds, females constituted a higher proportion (63%) of the students in the sample from urban schools, and of the three teachers involved, two were female. In contrast, males made up 59% of the students in the sample from rural schools, and of the five teachers involved, three were male. As we have already seen from the results

on the influence of teacher gender, girls perform better when their teacher is female, and this might be one reason for the results shown in Table 4. Secondly, and of greater relevance, there is the likelihood that boys in urban schools are more distracted from their studies than girls. To explore this, email interviews were conducted with three secondary school principals, seven secondary school teachers, and one undergraduate at the University of the South Pacific.

Of these respondents, seven were male and four female, and all were ni-Vanuatu with experience of living in either Port Vila or Luganville. The question asked of them was: "From your experience of living in towns, would you say that boys at Year 9/10 level are more distracted or more disengaged from their studies than girls? What are the things going on in town that distract boys more than girls?"

Of the 11 respondents, nine had no doubt that boys are more easily distracted than girls:

I think boys are much more distracted than girls. From my experience, girls generally are very committed in studying. Boys on the other hand are easily distracted by peers, which influences them to be involved in activities such as drinking, smoking tobacco and cigarettes at a very young age, going to nightclubs and some even creating gangs to steal in the shops and plenty more.

Teacher of English, Whitewood High School

Boys seemed to be more distracted than girls due to the following influences:

1. Peer pressure at school or from dropouts within their neighbourhoods. They would rather hang around with them doing drugs and drinking. This has sometimes led to gang fighting in schools as well as high absenteeism from schools. They come to school but spend the time hanging around school without attending classes.
2. Smoking marijuana and drinking alcohol.
3. Distractive entertainments such as clubbing, partying and ceremonial activities (weddings, deaths, etc.)
4. Parental neglect, which leads to a high amount of distractions especially with boys. This neglect could be due to parental divorce, single parenting or busy working parents who do not spend enough time

with the children, so creating a sense of loss in the children that makes them neglect school activities or lose interest in their education, with school becoming a pastime activity while transitioning to adulthood.

5. Distraction caused by social media (including Facebook) and the internet, which leads to a lot of unnecessary entertainment that can mislead most boys to think that life is all about entertainment. So when lessons are not entertaining the boys become bored and disengaged.

6. The school curriculum does not provide a wholesome education that caters for students' needs and abilities and does not provide the learning they need. The curriculum is mostly theoretical and not practical enough, so only a few will be interested.

Teacher of Social Science, Nagavika College

Boys are more distracted than girls. The biggest distraction today is the online video game PUBG (Players Unknown Battle Ground). Boys want to do little chores to earn themselves money for purchasing phone data. They play this game at home till late hours and even in school during breaks, lunch time or free periods. In addition to this we have TikTok, Facebook Madlpz and FruityLoops, Remix/Music Software, which are more suited to boys than girls.

Teacher of Social Science, Acacia Secondary School

Two respondents, however, had differing views:

No, girls are more easily distracted. Their distractions are drugs, boy-girl relationships, social media (especially Facebook), and physical appearance.

Deputy Principal, Namariu Junior Secondary School

Girls are more easily distracted because nowadays they usually have boy friends at an earlier age than in the past due to social platforms.

Teacher of Social Science, Banyan School

Table 5 summarizes respondents' perceptions of the main factors causing boys to be more easily distracted than girls from their studies at school in urban contexts.

In short, these interviews lend substantial support to the statistical finding that in urban schools, girls performed significantly better than boys in the Toolkit activity because of the many factors causing boys to become disengaged from formal education.

Table 5

Teachers' Perceptions of Factors Causing Boys to Have Greater Disengagement from School than Girls in Urban Areas of Vanuatu

Factor	Number of respondents reporting this factor (out of 11)
Taking alcoholic drinks	8
Smoking cigarettes/tobacco	6
Drinking kava	4
Taking marijuana/illegal drugs	4
Peer pressure, especially from those who have left school	4
Involvement in gangs	3
Going to night clubs	3
Entertainment provided on social media, internet, movies, etc.	3
Stealing	2
Boys have more freedom than girls	2
Mobile phones	2
Lack of parental care	2
Higher participation in sport than girls	1
Boy-girl relationships	1
Influence of swearing	1
Inadequate school curriculum	1
Online video game PUBG	1

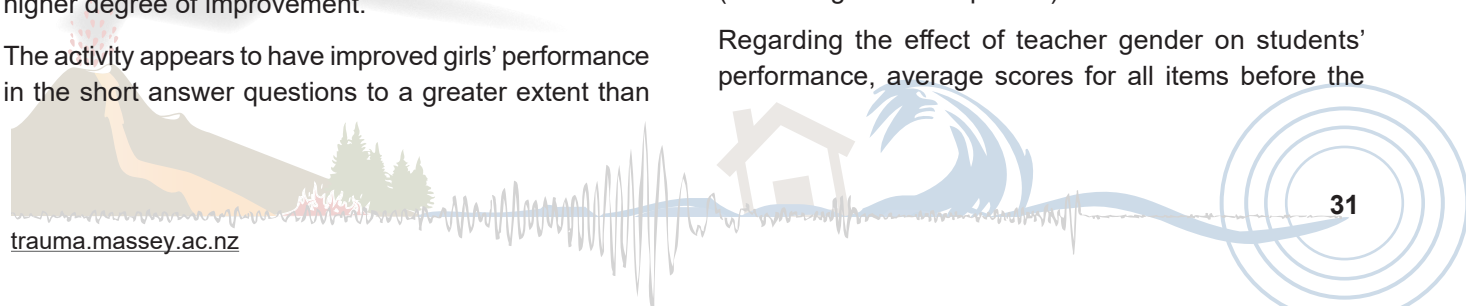
Discussion

This paper aims to examine the link between gender and climate change education through the use of the Climate Change Toolkit in Years 9-12. A sample of 112 female and 97 male students from eight classes in five secondary schools in Vanuatu participated in this discovery learning classroom activity. Measurement of changes in their knowledge, attitudes, and behaviour was obtained through the use of the same diagnostic questionnaire completed by respondents before and after undertaking the activity – a questionnaire comprising 20 agree/disagree items and seven open-ended short answer questions. Analysis of results shows that overall scores obtained for all items in the questionnaire were almost identical both before and after the intervention, with 0.49 for both genders beforehand and 0.81 for females and 0.80 for males afterwards. The same parity of performance was demonstrated by the magnitude of improvement between before and after scores. For knowledge, boys slightly outperformed girls in the agree/disagree items, while girls outperformed boys in the short answer questions. For attitudes, the Toolkit activity improved male scores more than female ones, while it was the reverse for behaviour, with females showing a higher degree of improvement.

The activity appears to have improved girls' performance in the short answer questions to a greater extent than

for boys. Six of these questions test knowledge about the nature and impacts of climate change, the enhanced greenhouse effect, ocean acidification, the role of trees in mitigating climate change, and anthropogenic causes of climate change – physical processes that boys in other situations are seen as grasping more readily than girls, as found by Kira and Komba (2015) and Ballew et al. (2018). If girls improved their scores more than boys in these six questions because of participating in the Toolkit activity, one inference is that the Toolkit itself was more helpful for girls in raising their scientific knowledge than it was for boys. Another inference is that it was equally effective for both sexes but that teenage girls had greater commitment to study (Hadjar et al, 2014; Jackman & Morrain-Webb, 2019; Van Houtte, 2004) and perhaps higher capacity to express their newly acquired knowledge in written form than did teenage boys. Similarly, the seventh short answer question, on intended behaviour, may have captured females' greater motivation to look for ways to care for their families and communities in the face of climate change and more intense disasters, in keeping with the findings of Zelezny et al. (2000) – a motivation stimulated by Toolkit pictures 9 and 10 (on hydro-meteorological disasters) and 11-16 (on strategies for adaptation).

Regarding the effect of teacher gender on students' performance, average scores for all items before the



intervention were significantly higher in those classes taught by female teachers than in those taught by males, but when students carried out the activity, those taught by male teachers then attained similar scores to those taught by females. One reason could be that male teachers were more effective in implementing the Toolkit activity. Another might be that the students taught by female teachers already knew more of the material under study. A third possibility is that the small sample size led to a distortion of results, and that with a much larger group of students and teachers there would have been no significant change in overall performance between students taught by male and female teachers.

When disaggregating results by domain of learning, however, a different pattern emerged. For both knowledge and attitudes, students in mixed classes taught by male teachers showed a greater improvement in scores than those taught by females. For behaviour, that pattern was reversed, with female-taught classes showing a more striking improvement in intention to adopt a greater variety of adaptation measures. This could be because female teachers have a heightened risk perception of environmental dangers (Bord & O'Connor, 1997; Xiao & McCright, 2012) and because of their natural tendency to stress the importance of caring for family and community (Zelezny et al., 2000).

For individual items, a similar reasoning could apply to the finding that female-taught classes achieved significantly higher scores than those taught by males for showing confidence in knowledge of climate change (item 19), in knowledge of the enhanced greenhouse effect and its causes (items 22 and 23), in explaining the role of trees in absorbing atmospheric carbon dioxide (item 25), in stating a range of climate change impacts in Vanuatu (item 26), and describing actions that bring greater resilience (item 27). Such results relate to McCright's (2010) view that women are more likely than men to have greater knowledge about climate change and to worry about its effects. As female teachers, therefore, this knowledge and worry may be communicated directly or indirectly in their classrooms, perhaps influencing students' performance in the Toolkit activity.

A significant finding has been that in terms of performance, the gender of the student and gender of the teacher are clearly linked, with both male and female teachers being more effective at teaching students of their own gender. The pattern for male teachers was the mirror image of that for female teachers: with male teachers, average scores for boys both before and after the intervention were higher than those for girls, while with female

teachers, girls had higher average scores than boys beforehand and afterwards. The findings support those of Muralidharan and Sheth (2016), who based their conclusions on 5 years of data from primary schools in the Indian state of Andhra Pradesh. However, they were looking at performance across all school subjects, and at levels corresponding to Years 1 to 6 in Vanuatu's educational system, while this study focuses on just one classroom activity carried out by an older cohort of mostly Year 9-10 students.

In the final aspect of this paper's investigations - differences in performance by gender between urban and rural schools - another clear pattern emerged, with females achieving statistically significant higher scores than males when the Toolkit activity was conducted in urban schools, while there was an apparent (but non-significant) trend of males outperforming females in rural schools. One reason for this pattern could be that the girls and female teachers accounted for the majority of the sample in urban schools, and results have already demonstrated that girls perform better when their teacher is female. On the other hand, boys and male teachers made up a higher proportion of the sample in rural schools. A second reason comes from my own observations through 35 years of teaching in Vanuatu that distractions in urban areas have a greater influence on boys than girls. I therefore decided to conduct email interviews with 10 teachers and one university student with experience of urban living in Vanuatu and triangulate their views with the quantitative data collected from students.

Of the 11 respondents, nine confirmed that in an urban environment, Year 9/10 boys are more easily distracted from their studies than girls. The most common factor involved was perceived as the consumption of alcoholic drinks, closely followed by smoking cigarettes and alcohol. Other important influences are the consumption of kava, marijuana, and other illegal drugs, usually associated with peer pressure from boys in the neighbourhood who have already dropped out of school. In some cases, membership in a gang becomes all-consuming and involves a boy in stealing and going to night-clubs. In general, respondents said that it is customary for teenage boys to be accorded greater freedom than girls, with the latter under stricter parental control and required to return home straight after school to assist with household duties and continue with their studies. The influence of social media was seen to affect both boys and girls in urban areas, but was of less significance in rural schools, where internet access may not be readily available.

These findings are in line with research in Caribbean schools by Jackman and Morrain-Webb (2019). They also support the findings of Hadjar et al. (2014) that boys are more alienated from school than girls, leading to “non-conformity and anti-social behaviour in the school environment” (p. 121) and lower educational achievement. This study drew on literature in a European context but is relevant to Vanuatu in that its two urban centres will offer a similar intensity of distractions to that found in a Western environment – which in turn will affect boys more than girls and perhaps explain girls’ superior performance in the Toolkit activity when undertaken in towns.

On a broader note, this study has implications for building the resilience of Vanuatu’s young people to climate change and disasters. The nation’s two key policy documents on resilience – the Vanuatu Climate Change and Disaster Risk Reduction Policy 2016-2030 (VCCDRRP) (Government of Vanuatu and SPC, 2015) and Vanuatu 2030: The National Sustainable Development Plan (Government of Vanuatu, 2016) – refer in general terms to the importance of educating students about resilience issues. Environmental goal # 3.4 of the VCCDRRP aims to “promote and ensure strengthened resilience and adaptive capacity to climate related, natural and man-made hazards” (p. 14). The VCCDRRP says that climate change and disaster risk reduction have been mainstreamed into the National Curriculum Statement. However, the implementation of revised curricula since 2010 has been very slow, and for junior secondary school students, the Toolkit was in 2022 one of the few resources that can promote their resilience. This paper has demonstrated that both girls and boys at junior secondary school level benefit from exposure to the Toolkit and associated activity, with average overall scores rising for both boys and girls, and a marked improvement in scores for behaviour intended to bolster adaptation, particularly in girls.

The impact on girls’ knowledge, attitudes, and behaviour, especially when their teacher is female, may perhaps contribute towards the VCCDRRP’s goal of greater empowerment for women in climate change issues:

Climate change and disaster risks impact greatly on the lives of women and their social and economic wellbeing. In disaster preparedness, as well as response and recovery, women make enormous contributions to the welfare of their communities. It is vital that women have full opportunities to participate in policy development, decision-making

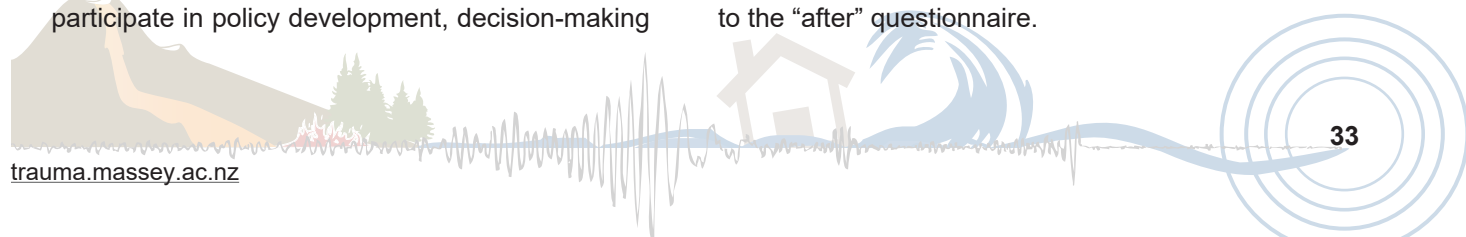
and implementation at all levels. Initiatives are under way to ensure that gender considerations are mainstreamed into all climate change and disaster risk reduction efforts, and efforts should be increased. Women have historically been excluded from participating on committees and other decision-making forums, and their full representation needs to be assured.

(Government of Vanuatu & SPC, 2015, Section 8.2, p. 26)

In the long-term, however, the usefulness of the CC Toolkit may decline as new syllabi are introduced at junior secondary level which reflect the UN’s Sustainable Development Goals as well as the VCCDRRP and Vanuatu’s National Sustainable Development Plan. In this way, students and teachers should have access to a richer range of pedagogy and materials that help to build “a stable, sustainable and prosperous Vanuatu” (Government of Vanuatu, 2016, p. 1) and ensure greater resilience to climate change for the country’s youth.

Limitations of the Research

This study on the effectiveness of the Climate Change Toolkit on male and female resilience to climate change and disasters has several limitations. Firstly, the sample included only 209 students from eight classes. The larger parent sample of 363 students and 19 classes from which information was collected to assess the influence of other variables, although representative of males and females and urban and rural schools, French- and English-medium classes, and a range of five year levels, could not be used for gender-specific data because this had not been planned at the outset of the research. Secondly, investigations were carried out during a period of stress and uncertainty for teachers and students because of the state of emergency arising from COVID-19 and, initially, Cyclone Harold. Schools were closed during the first few months of this period, and then when they did resume, teachers did not prioritize the task of undertaking this classroom intervention on climate change. Restrictions on international travel also imposed limitations on the researcher’s ability to interact with teachers and students involved. Thirdly, it is likely that the activity was not always undertaken in the prescribed manner, because of shortage of time. In particular, the learning that would occur through class discussion of the pictures once the activity had been completed may not in fact have taken place in several classrooms. This would impact upon the accuracy of students’ answers to the “after” questionnaire.



Conclusion

As part of my overall research into the effectiveness of formal learning about climate and disaster resilience in Vanuatu, I examined the situation at the junior secondary level of education, focusing on the usefulness of "Learning About Climate Change the Pacific Way" and its accompanying discovery learning activity. As the research unfolded, it became possible to make comparisons between the performance of girls and boys, and between girls and boys in rural and urban settings. Evidence from questionnaires completed by a sample of 112 girls and 97 boys from eight classes, drawn largely from Years 9 and 10, demonstrates that when average scores for knowledge, attitudes, and behaviours were aggregated, there was no significant difference in performance between males and females either before or after the intervention. On the other hand, when the gender of the teacher was considered, female students performed better overall when their teacher was female and male students achieved higher scores when their teacher was male. A closer examination of the situation in urban and rural schools confirmed this same trend: in urban schools, female students outperformed males both before and after the intervention, while in rural schools males outperformed females, although to a lesser extent and not statistically significantly.

This data was collected during the first 15 months of the COVID-19 pandemic, with the researcher having to collect information by remote means. This was a particular challenge when endeavouring to evaluate reasons for the differences observed between male and female performance, with email interviews having to replace face-to-face contact with teachers and students. Nevertheless, the emerging evidence suggests that the distractions of living in an urban area exert a greater influence on boys than on girls, with a consequent tendency for urban boys to become more disengaged from formal education, mainly because of the influence of alcohol, cigarettes, kava, marijuana, peer pressure from those who have already left school, greater personal freedom, and accessibility to the internet.

Teachers and students said that they appreciated the manner in which the cooperative and discovery learning strategies of this intervention engaged their students, dealt with real-life situations in a Pacific Island context and appeared to have an impact on students' knowledge, attitudes, and intended patterns of behaviour. Yet teachers made no comments on students' misunderstandings of mitigation and adaption nor of scientific processes involved in the enhanced

greenhouse effect or the absorption of carbon dioxide by forests and oceans.

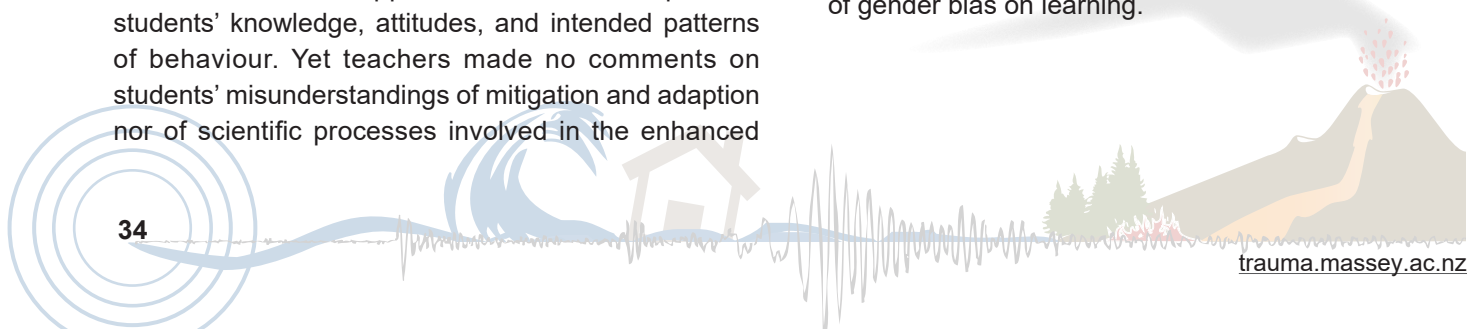
A tentative conclusion to draw from the findings is that while the Toolkit had a similar effect on the overall performance of both genders, cultural factors in Vanuatu suggest that girls' resilience to climate change will be better when their teacher is female, and boys' resilience will improve when the teacher is male. Additionally, boys' interest in becoming resilient to climate change is more likely to occur when they are being educated in a rural environment that has fewer distractions than those of the city and can more readily observe the changes in the environment that are already resulting from disasters and the climate crisis. Although this study was specific to Vanuatu, some of the findings may have implications for other Pacific Island nations.

Finally, this study demonstrates the usefulness of the Climate Change Toolkit in contributing towards the broad aims of Vanuatu's two key policies on CCDRR and sustainable development. It has raised the competency of boys and especially of girls to gain more knowledge of the scientific aspects of climate change, have greater concern over its impacts, be more willing to help their community prepare for climate change and disasters, and to undertake actions that will forge greater adaptive capacity. In time, its usefulness may lessen as new syllabi and materials become available, but for the moment, it is a proven educational resource for Vanuatu.

Recommendations

It is recommended that further studies be done on how the performance of students in the Climate Change Toolkit activity may be affected by gender of student and gender of teacher, but that such studies involve a much larger cohort of students and teachers and include a more systematic element of initial teacher training. This has cost implications for bringing teachers together in a central location in each of Vanuatu's provinces, and also for the provision of picture sets and updated teacher guides.

There may also be a further, wider issue that needs to be addressed: how to effect changes in Vanuatu culture whereby a person's gender does not define the way in which he/she acquires or imparts knowledge, skills, attitudes, or desired behaviours, so removing the effect of gender bias on learning.



Disclosure statement

The author was the Vanuatu contributor to the set of pictures Learning about Climate Change the Pacific Way during its development in 2012-2014. He has 35 years of teaching experience in educational institutions in Vanuatu.

Funding

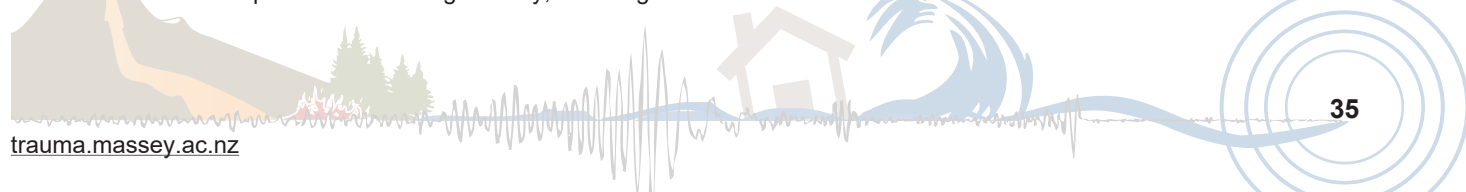
This article was produced on a purely voluntary basis. No funding was received from any source, neither from within Vanuatu nor from overseas donor partners.

Acknowledgement

I wish to acknowledge the guidance and support of Dr Phil Wood, of Bishop Grosseteste University, Lincoln, UK, in producing this article.

References

- Ballew, M., Marlon, J., Leiserowitz, A., & Maiback, E. (2018). *Climate Note: Gender differences in public understanding of climate change*. Yale Programme on Climate Change Communication. <https://climatecommunication.yale.edu/publications/gender-differences-in-public-understanding-of-climate-change/>
- Bergland, C. (2013). Scientists identify why girls often mature faster than boys. *Psychology Today*. www.psychologytoday.com/au/blog/the-athletes-way/201312/scientists-identify-why-girls-often-mature-faster-boys#:~:text=The%20researchers%20conclude%20that%20this,%2C%202013%2C%20in%20Cerebral%20Cortex
- Bord, R., & O'Connor, R. (1997). The gender gap in environmental attitudes: The case of perceived vulnerability to risk. *Social Science Quarterly*, 78(4), 830–840.
- Brody, S., Zahran, S., Vedlitz, A., & Grover, H. (2008). Examining the relationship between physical vulnerability and public perceptions of global climate change in the United States. *Environment and Behavior*, 40(1), 72–95. <https://doi.org/10.1177/0013916506298800>
- Brouwer, R., Schutte, J., Janssen, R., Boomsma, D., Hulshoff Pol, H., & Schnack, H. (2021). The speed of development of adolescent brain age depends on sex and is genetically determined. *Cerebral Cortex*, 31(2), 1296–1306. <https://doi.org/10.1093/cercor/bhaa296>
- Bündnis Entwicklung Hilft. (2021). *World Risk Report 2021*. <https://reliefweb.int/sites/reliefweb.int/files/resources/2021-world-risk-report.pdf>
- Connell, J. (2013). *Islands at risk? Environments, economies and contemporary change*. Edward Elgar Publishing Ltd.
- DeCarlo, M. (2018). Sampling in qualitative research. In *Scientific inquiry in social work*. PressBooks.
- Figueroa, M. (2000). Making sense of male experience: The case of academic underachievement in the English-speaking Caribbean. *IDS Bulletin*, 31(2), 68–74. <https://doi.org/10.1111/j.1759-5436.2000.mp31002009.x>
- Government of the Republic of Vanuatu. (2016). *Vanuatu 2030 - The People's Plan: National Sustainable Development Plan 2016-2030*. Department of Strategic Policy, Planning and Aid Coordination. www.greengrowthknowledge.org/national-documents/vanuatu-2030-peoples-plan-national-sustainable-development-plan-2016-2030
- Government of the Republic of Vanuatu, & Secretariat of the Pacific Community (SPC). (2015). *Vanuatu Climate Change and Disaster Risk Reduction Policy (VCCDRRP) 2016-2030*. www.preventionweb.net/publication/vanuatu-climate-change-and-disaster-risk-reduction-policy-2016-2030
- Hadjar, A., Krolak-Schwerdt, S., Priem, K., & Glock, S. (2014). Gender and educational achievement. *Educational Research*, 56(2), 117–125. <https://doi.org/10.1080/0013181.2014.898908>
- Hamilton, L. (2011). Education, politics, and opinions about climate change evidence for interaction effects. *Climatic Change*, 104, 231–242. <https://doi.org/10.1007/s10584-010-9957-8>
- Hemstock, S., Des Combes, H., Martin, T., Vaike, F., Maitava, K., Buliruarua, L-A., Satiki, V., Kua, N., & Marawa, T. (2017). A case for formal education in the Technical, Vocational Education and Training (TVET) sector for climate change adaptation and disaster risk reduction in the Pacific Islands region. In W. Leal Filho (Ed.), *Climate change adaptation in Pacific countries: Fostering resilience and improving the quality of life* (pp. 309–324). Springer. <https://doi.org/10.1007/978-3-319-50094-2>
- Hornsey, M., Harris, E., Bain, P., & Fielding, K. (2016). Meta-analyses of the determinants and outcomes of belief in climate change. *Nature Climate Change*, 6(6), 622–626. <https://doi.org/10.1038/nclimate2943>
- Hyde, J. (2005). The gender similarities hypothesis. *American Psychologist*, 60(6), 581–592. <https://doi.org/10.1037/0003-066X.60.6.581>
- Jackman, W., & Morrain-Webb, J. (2019). Exploring gender differences in achievement through student voice: Critical insights and analyses. *Cogent Education*, 6(1). <https://doi.org/10.1080/2331186X.2019.1567895>
- Jha, J., & Kelleher, F. (2006). *Boys' underachievement in education: An exploration in selected Commonwealth countries*. Commonwealth Secretariat and Commonwealth of Learning. <http://oasis.col.org/bitstream/handle/11599/168/BoysUnderachievement.pdf?sequence=1>
- Joseph, S. (2016). Re-masculinization of the teaching profession: Towards a policy of attracting, recruiting and retaining male teachers in the Trinidad and Tobago school system. *Advances in Social Science Research Journal*, 3(3), 123–132. <https://doi.org/10.14738/assrj.33.1891>
- Kagawa, F., & Selby, D. (2012). Ready for the storm: Education for disaster risk reduction and climate change adaptation and mitigation. *Journal of Education for Sustainable Development*, 6 (2) 207–217. <https://doi.org/10.1177/0973408212475199>
- Kira, E., & Komba, S. (2015). Comparison between the understanding levels of boys and girls on the concepts of environmental degradation, meteorology and climate change in Tanzanian secondary schools. *Journal of Education and E-Learning Research*, 2(4), 79–86.
- Le Dé, L., Rey, T., Leone, F., & Gilbert, D. (2018). Sustainable livelihoods and effectiveness of disaster responses: A case study of tropical cyclone Pam in Vanuatu. *Natural Hazards*, 91, 1203–1221. <https://doi.org/10.1007/s11069-018-3174-6>
- Lim, S., Han, C., Uhlaas, P., & Kaiser, K. (2013). Preferential detachment during human brain development: Age- and sex-specific structural connectivity in Diffusion Tensor



- Imaging (DTI) data. *Cerebral Cortex*, 25(6), 1477-1489. <https://doi.org/10.1093/cercor/bht333>
- Mahuri, D. (2019). *Teachers' perceptions of socio-cultural practices on students' academic achievement in North Pentecost, Vanuatu*. [Master's thesis, University of the South Pacific, Suva, Fiji].
- Marshall, M. (1996). Sampling for qualitative research. *Family Practice*, 13(6), 522-525. <https://doi.org/10.1093/fampra/13.6.522>.
- McCright, A. (2010). The effects of gender on climate change knowledge and concern in the American public. *Population and Environment*, 32, 66–87. <https://doi.org/10.1007/s11111-010-0113-1>
- McCright, A., Dunlap, R., & Xiao, C. (2013). Perceived scientific agreement and support for government action on climate change in the USA. *Climatic Change*, 119, 511–518. <https://doi.org/10.1007/s10584-013-0704-9>
- McCright, A., & Sundström, A. (2013). Examining gender differences in environmental concern in the Swedish general public, 1990–2011. *International Journal of Sociology*, 43(4), 63–86. <https://doi.org/10.2753/IJS0020-7659430402>
- Ministry of Education. (2010). *Vanuatu National Curriculum Statement, 2010*. Government of Vanuatu.
- Ministry of Education and Training. (2021). *Education Statistics: Basic tables of 2020*. Government of Vanuatu. https://moet.gov.vu/docs/statistics/2020%20MoET%20Education%20Statistics%20Report_2020.pdf
- Ministry of Education and Training. (2022). *Education Statistics: Basic tables of 2021*. Government of Vanuatu. https://education.gov.vu/docs/statistics/2021%20MoET%20STATISTICAL%20REPORT_2021.pdf
- Muralidharan, K., & Sheth, K. (2016). Bridging education gender gaps in developing countries: The role of female teachers. *Journal of Human Resources* 51(2), 269-297. <https://doi.org/10.3368/jhr.51.2.0813-5901R1>
- Nakaseko, E., Kotera, S., & Nakazawa, M. (2022). Smoking and drinking behaviour, knowledge and attitudes among urban and rural public-school students in Efate island, Vanuatu: A comparative study. *Archives of Public Health* 80, 1-9. <https://doi.org/10.1186/s13690-022-00929-9>
- Nnamani, S., & Oyibe, O. (2016). Gender and academic achievement of secondary school students in Social Studies in Abakaliki Urban of Ebonyi State. *British Journal of Education*, 4(8), 84-95.
- Pearson, A., Ballew, M., Naiman, S., & Schuldt, J. (2017, April 26). Race, class, gender and climate change communication. *Oxford Research Encyclopedia of Climate Science*. <https://doi.org/10.1093/acrefore/9780190228620.013.412>
- Pierce, C. (2019). Realities of teaching climate change in a Pacific island nation. In W. Leal Filho & S. Hemstock (Eds.), *Climate change and the role of education: Climate change management* (pp. 319-347). Springer Nature. https://doi.org/10.1007/978-3-030-32898-6_18
- Pierce, C. (2023). *Effectiveness of formal and traditional learning about climate and disaster resilience in Vanuatu*. [Doctoral dissertation, University of Leicester]. <https://doi.org/10.25392/leicester.data.23180306>
- Secretariat of the Pacific Community, & Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH. (2013). *Learning about climate change the Pacific way: A guide for Pacific teachers – Vanuatu*. www.pacificclimatechange.net/sites/default/files/documents/Vanuatu_2014_LaCCpw_Teacher_Guide.pdf
- Secretariat of the Pacific Community, & Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH. (2014). *Learning about climate change the Pacific way: A visual guide – Vanuatu*. www.pacificclimatechange.net/document/learning-about-climate-change-pacific-way-visual-guide-vanuatu
- Stanford Medicine Children's Health. (2024). *Cognitive development in the teen years*. www.stanfordchildrens.org/en/topic/default?id=cognitive-development-in-adolescence-90-P01594
- Stevenson, R., Nicholls, J., & Whitehouse, H. (2017). What is climate change education? *Curriculum Perspectives*, 37, 67-71. <https://doi.org/10.1007/s41297-017-0015-9>
- Ullah, R., & Ullah, H. (2019). Boys versus girls' educational performance: Empirical evidences from global north and global south. *African Educational Research Journal*, 7(4), 163-167. <https://doi.org/10.30918/AERJ.74.19.036>
- United Nations Department of Economic and Social Affairs. (2015). *Sustainable Development: The 17 Goals*. <https://sdgs.un.org/goals>
- United Nations Educational, Scientific and Cultural Organisation. (2015). *Not just hot air: Putting climate change education into practice*. www.uncclearn.org/wp-content/uploads/library/unesco01_0.pdf
- Van der Linden, S. (2015). The social-psychological determinants of climate change risk perceptions: Towards a comprehensive model. *Journal of Environmental Psychology*, 41, 112–124. <https://doi.org/10.1016/j.jenvp.2014.11.012>
- Van Houtte, M. (2004). Gender context of the school and study culture, or how the presence of girls affects the achievement of boys. *Educational Studies*, 30(4), 409-423. <https://doi.org/10.1080/0305569042000310336>
- Vanuatu National Statistics Office. (2021). *2020 National Population and Housing Census: Basic tables report, volume 1*. Government of Vanuatu. https://sdd.spc.int/digital_library/vanuatu-2020-national-population-and-housing-census-basic-tables-report-volume-1
- Walid, M. (2017). Towards a mega-Pacific island education curriculum for climate adaptation, blending traditional knowledge in modern curriculum. In W. Leal Filho (Ed.), *Climate change adaptation in Pacific countries: Fostering resilience and improving the quality of life* (pp. 271-285). Springer. https://doi.org/10.1007/978-3-319-50094-2_16
- Walshe, R., & Nunn, P. (2012). Integration of indigenous knowledge and disaster risk reduction: A case study from Baie Martelli, Pentecost Island, Vanuatu. *International Journal of Disaster Risk Science*, 3, 185–194. <https://doi.org/10.1007/s13753-012-0019-x>
- Xiao, C., & McCright, A. (2012). Explaining gender differences in concern about environmental problems in the United States. *Society & Natural Resources*, 25(11), 1067–1084. <https://doi.org/10.1080/08941920.2011.651191>
- Zelezny, L., Chua, P., & Aldrich, C. (2000). New ways of thinking about environmentalism: Elaborating on gender differences in environmentalism. *Journal of Social Issues*, 56(3), 443-457. <https://doi.org/10.1111/0022-4537.00177>

Appendix

Form for tabulating numerical scores for responses

Name of school and class:

Date:

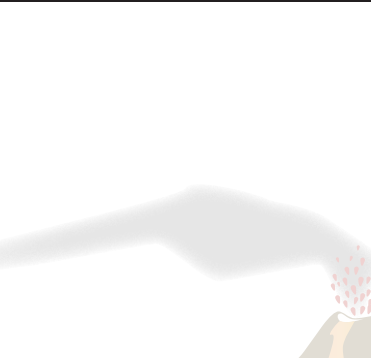
Number of students participating (n):

Name of teacher:

Questions asked before or after use of Toolkit pictures? (Encircle): **BEFORE** AFTER

	Statement	+ 1 (correct) - 1 (incorrect)	My opinion				
			Agree	Disagree	Don't know	Total score(T)	Av score (T ÷ n)
1.	1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.						
2.	2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.						
3.	3. Our climate changes because of both natural and human factors						
4.	4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.						
5.	5. Climate variability means that the climate of a place may change from year to year.						
6.	6. During an El Niño period, Vanuatu experiences droughts.						
7.	7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.						
8.	8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.						
9.	9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.						
10.	10. Carbon dioxide is absorbed by forests and the oceans						
11.	11. Coral reefs are being damaged by warmer temperatures and ocean acidification.						
12.	12. Vanuatu's greatest climatic dangers are cyclones and droughts.						
13.	13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.						
14.	14. Using compost is a sustainable form of gardening.						
15.	15. Catching fish with nets that have very small holes is a sustainable form of fishing.						
16.	16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.						
17.	17. I am worried that climate change will bring great dangers to the world in the future.						
18.	18. I want to join a student strike to show my concern about climate change.						
19.	19. I know a lot about the causes and impacts of climate change.						
20.	20. I want to help my community prepare for climate change and disasters.						

	Question	Answers
21.	What does "climate change" mean? 5: Accurate explanation of the concept and processes involved (e.g. significant <u>long-term change</u> in global climate or average weather patterns) 3: Satisfactory level of understanding of the processes involved (e.g. change in climate or change in average weather conditions or patterns) 1: Limited level of understanding (e.g. change over a period of time; it is getting hotter; there are longer droughts) 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	(Student answers are recorded here, then a score of 0-5 given to each one and all scores totalled for the question)

22.	What is the enhanced greenhouse effect? 5: Accurate explanation of the concept and processes involved (e.g. an <u>increase in greenhouse gases</u> in the atmosphere due to <u>human activities</u>, which leads to <u>greater atmospheric heating</u>) 3: Satisfactory level of understanding of the processes involved (e.g. an increase in dangerous gases in the atmosphere; increase in gases that absorb/trap heat radiated back from the earth) 1: Limited level of understanding (e.g. gases that cause global warming; gases absorbing the heat radiated back from earth; effects caused by GHG such as CO₂, methane, etc; effects such as melting ice, sea level rise, destruction of plants and animals, warmer temperatures) 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
23.	What are three human activities that are putting extra greenhouse gases into the atmosphere, so causing the enhanced greenhouse effect? Valid activities are: • Burning of fossil fuels in vehicles/ships/ aircraft • Burning of fossil fuels in industry/ factories • Deforestation • Cattle rearing • Rice cultivation. • Garbage/rubbish tips, burning of rubbish/plastic 5: Accurate: Three valid activities identified 3: Satisfactory: Two valid activities identified 1: Limited: One valid activity identified 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
24.	What causes ocean acidification? 5: Accurate explanation of the process involved (e.g. absorption of more and more CO₂ by the oceans; oceans acting as carbon sinks; absorption of CO₂ by oceans causes weak carbonic acid; increase of CO₂ in the oceans, which lowers pH value) 3: Satisfactory level of understanding of the processes involved (e.g. carbon dioxide is absorbed by the oceans) 1: Limited level of understanding (e.g. carbon dioxide; carbonic acid; methane, ozone, CO₂; warmer temperatures; pollution by oil/fuel/chemicals/sewage) 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
25.	Why is forestry (planting and caring for trees) important for mitigating climate change? 5: Accurate explanation of the concepts and processes involved (e.g. trees absorb carbon dioxide from the atmosphere, so reducing atmospheric CO₂ content, so slowing/ lessening global warming and climate change.) 3: Satisfactory level of understanding of the processes involved (e.g. trees absorb carbon dioxide [and give out oxygen]; trees reduce the amount of CO₂ in the atmosphere) 1: Limited level of understanding (e.g. adaptation measures: reduce soil erosion; provide habitat for living organisms; provide	

	<i>shade; provide materials for house construction and medicines; protect the environment; support livelihoods, etc.)</i> 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
26.	What are some of the main impacts of climate change in Vanuatu? <i>Valid impacts are:</i> • Sea level rise • Coastal erosion • Damage to coral reefs/aquatic life • Ocean acidification • Longer periods of drought • Loss of crops due to drought or higher temperatures • Higher temperatures / hot temperatures • More extreme weather events, e.g. stronger cyclones, more floods • Change in fruiting/flowering seasons of trees and plants • Other, e.g. distress, increase in diseases such as malaria and dengue 5: Accurate: At least 3 valid impacts 3: Satisfactory: Two valid impacts 1: Limited: One valid impact 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	
27.	What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters? List 3. <i>Answer should give adaptation strategies appropriate at family or community level for preparing for CC / more intense disasters, e.g. plant more trees/reafforestation, build stronger houses, get larger water tanks, re-locate buildings away from the shoreline or a river, plant mangroves, use composting and mulching, use solar energy/panels, plant and eat more aelan kakae, plant crop varieties that are resilient to CC and disasters, encourage more gardening, establish community conservation areas/ marine conservation areas, use environmentally-friendly fishing methods (e.g. canoes), conserve/save water, conduct CC awareness, encourage greater unity in the community. Mitigation measures can be mentioned, e.g. use of renewable resources, use public transport rather than private vehicles, recycling, avoid burning rubbish/plastic, etc. Preparations for disasters can also be given, e.g. food preservation, gathering food and water, cyclone-proof housing, etc.</i> 5: Accurate: Three valid actions identified 3: Satisfactory: Two valid actions identified 1: Limited: One valid action identified 0: Irrelevant, unclear, don't know TOTAL SCORE: AVERAGE SCORE: ÷ =	

Example of a summary data form for one className of school and class: Sandalwood JSS Yrs 9/10 Science Date: 22nd March 2021

Number of students participating (n): 21 Name of teacher:

Questions asked before or after use of Toolkit pictures? BEFORE AFTER

	Statement + 1 (correct)  - 1 (incorrect) 	Agree	Disagree	Don't know	Total score (T)	Av score (T ÷ n)
1.	1. Weather refers to the way that atmospheric conditions such as temperature, rainfall, pressure and wind are always changing.	14	2	5	12	0.57
2.	2. Climate refers to average conditions of temperature, rainfall, etc. over a long period of time.	16	2	3	14	0.67
3.	3. Our climate changes because of both natural and human factors	9	5	7	4	0.19
4.	4. Evaporation occurs when water vapour in the air changes back to tiny droplets of water.	3	10	8	7	0.33
5.	5. Climate variability means that the climate of a place may change from year to year.	8	5	8	3	0.14
6.	6. During an El Niño period, Vanuatu experiences droughts.	11	3	7	8	0.38
7.	7. The greenhouse effect means that certain gases in the atmosphere absorb the heat being radiated back from the Earth, so keeping the atmosphere warm.	7	6	8	1	0.05
8.	8. Greenhouse gases include carbon dioxide, methane, nitrous oxide and water vapour.	9	4	8	5	0.24
9.	9. For the past 200 years, human activities are putting extra greenhouse gases into the atmosphere.	4	5	12	-1	-0.05
10.	10. Carbon dioxide is absorbed by forests and the oceans	9	5	7	4	0.19
11.	11. Coral reefs are being damaged by warmer temperatures and ocean acidification.	6	2	13	4	0.19
12.	12. Vanuatu's greatest climatic dangers are cyclones and droughts.	5	7	9	-2	-0.10
13.	13. Adaptation refers to the actions we can take to reduce the levels of greenhouse gases in the atmosphere.	4	4	13	0	0.00
14.	14. Using compost is a sustainable form of gardening.	8	6	7	2	0.10
15.	15. Catching fish with nets that have very small holes is a sustainable form of fishing.	4	10	7	6	0.29
16.	16. Vanuatu is one of the most vulnerable countries in the world to natural disasters.	8	5	8	3	0.14

Knowledge

17.	17. I am worried that climate change will bring great dangers to the world in the future.	8	6	7	2	0.10
18.	18. I want to join a student strike to show my concern about climate change.	7	6	8	7	0.33
19.	19. I know a lot about the causes and impacts of climate change.	3	10	8	-7	-0.33
20.	20. I want to help my community prepare for climate change and disasters.	6	7	8	-1	-0.05

Attitudes

21.	What does "climate change" mean?	2	0.10
22.	What is the enhanced greenhouse effect?	6	0.29
23.	What are 3 human activities putting extra GHGs into the atmosphere?	16	0.76
24.	What causes ocean acidification?	5	0.24
25.	Why is forestry important for mitigating climate change?	19	0.90
26.	What are some of the main impacts of climate change in Vanuatu?	14	0.67

Knowledge

27.	What kinds of actions can I take to help my family and community become better prepared for the impacts of climate change and more intense disasters? List 3.	27	1.29
-----	---	----	------

Behaviour